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MILITARY SPECIFICATION

LEVELS, SURVEYING, SELF-LEVELING AND TILTING

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers internal-focusing, inverted or erect image, self-leveling, and tilting type leveling instruments, with tripods and accessories.

1.2 Classification. The levels shall be of the following types, styles, and categories, as specified (see 6.2):

Type I - Tilting level.
Style A - With micrometer eyepiece and plane-parallel attachment.
Style B - Without optional micrometers and plane-parallel attachment.
Type II - Self-leveling level.
Style A - Without horizontal circle.
Style B - With horizontal circle for stadia and simple surveys.
Category 1 - 1st order accuracy.
Category 2 - 2d order accuracy.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-C-675	- Coating of Glass Optical Elements (Anti-Reflection).
MIL-T-704	- Treatment and Painting of Materiel.

FSC 6675

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Mobility Equipment Research and Development Command, ATTN: DRXFB-DS, Fort Belvoir, VA 22060 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-S-3534	- Surveying Instruments and Accessories, Packaging of.
MIL-T-14091	- Tripod, Surveying: Extension-Leg-Type.

STANDARDS

MILITARY

MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-731	- Quality of Wood Members for Containers and Pallets.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

NATIONAL BUREAU OF STANDARDS

Handbook H28 - Screw-Thread Standards for Federal Services.
Special Publication 374 - Method for Determining the Resolving Power of Photographic Lenses.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, ATTN: Tariff Publishing Officer, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Associations, Inc., ATTN: Tariff Order Section, 1616 P Street NW, Washington, DC 20036).

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PRINCETON STANDARD TEST FOR ESTIMATING THE ACCURACY OF A LEVEL.

(Application for copies should be addressed to the Commander, U.S. Army Mobility Equipment Research and Development Command (DRXFB-RE), Fort Belvoir, VA 22060.)

3. REQUIREMENTS

3.1 Description.

3.1.1 General. The level shall consist of an internal focusing telescope, sunshade, level base adaptable to the universal tripod, leveling screws, image lens system, universal tripod, accessories and carrying case. The parts of any specific type and style level shall be interchangeable with the corresponding parts of any level of the same designation.

3.1.2 Type I levels. The Type I level shall be the tilting type, equipped with a graduated differential tilting screw and an enclosed tubular level vial with prism assembly and eyepiece to enable a split image view of the tubular vial. The level shall include a circular spirit level for preliminary leveling.

3.1.2.1 Type I, Style A. Type I, Style A levels shall be equipped with an optical micrometer attached to a plane-parallel glass plate mounted in front of the objective lens.

3.1.2.2 Type I, Style B. Type I, Style B levels shall not have plane-parallel attachment or an optical mircometer.

3.1.3 Type II levels. Type II levels shall be equipped with a pendulum compensator for self-leveling capabilities. When specified (see 6.2), the level shall be equipped with an optical micrometer attached to a plane-parallel glass plate mounted in front of the objective lens. Type II levels may have either an inverted or erect image, as specified (see 6.2).

3.1.3.1 Type II, Style A. Type II, Style A levels shall be without a horizontal circle.

3.1.3.2 Type II, Style B. Type II, Style B levels shall be equipped with a horizontal circle.

3.2 Qualification. Levels furnished under this specification shall be products which are qualified for listing on the applicable qualified products list (see 4.3 and 6.3).

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3.3 Initial production. When specified (see 6.2), the contractor shall furnish to the Government one or more levels (see 6.2) for inspection as specified in 4.4.

3.4 Material. Material shall be as specified herein. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification. Materials shall be nonmagnetic and shall be corrosion-resistant or treated to resist corrosion. Material of all bearing surfaces in contact with each other shall have the same coefficient of expansion; if not, provision shall be made for differential movement. Materials shall be free of defects and imperfections that could contribute to premature failure of the level, or reduce or impair the usability of the unit.

3.5 Environmental.

3.5.1 Operational temperature. The level, tripod, cases, and accessories shall operate as specified herein in any ambient temperature from plus 120° F for not less than 4 hours, to minus 25° F for not less than 6 hours. All mechanical movements shall remain free and shall function satisfactorily without need of readjustment or change of lubricant.

3.5.2 Storage and transit temperature conditions. The level, tripod, cases, and accessories shall not be damaged as a result of storage under the following conditions:

Air temperature:
High - Plus 155° F; time period: 4 hours.
Low - Minus 50° F; time period: 12 hours.

3.5.3 Rainfall. When in the operating position, the level shall not be damaged mechanically or optically and shall show no indication of moisture penetration after exposure to rainfall of 3 inches, plus or minus 1 inch, per hour for 5 minutes. The closed level carrying case and transport case shall show no indication of moisture penetration after exposure to rainfall of 3 inches, plus or minus 1 inch, per hour for a period of 1 hour.

3.5.4 Sand and dust. When in the operating position, the instrument shall not be damaged mechanically or optically and shall operate at the specified accuracy when subjected to sand and dust blown by a 100- to 500-foot-per-minute wind for 15 minutes. The level carrying case and transport case shall show no indication of sand or dust penetration after exposure to sand and dust blown by a 100- to 500-foot-per-minute wind for 1 hour.

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3.5.5 Shock and vibration. The instrument and accessories shall be constructed so that no fixed part shall become loose, no movable part or permanently set adjustment shall shift its setting or position, and no degradation shall be caused in the performance specified herein when subjected to the shock test specified in 4.6.2.20 and the vibration test specified in 4.6.2.19.

3.6 Performance. When tested as specified in 4.6.2.3, the instrument shall perform in the accuracy catagory as specified (see 1.2).

3.7 Telescope. The telescope shall be internal-focusing, completely encased, and provided with a sunshade. The focusing knob shall be located on the top or side. A nonrecessed, wide frame finder used for quick sighting of the objective shall be located on the telescope.

3.7.1 Optical system. The optical system shall be free from spherical and chromatic aberration, internal reflection, astigmatism, coma, curvature, and distortion. A clear, sharply defined image shall be provided throughout the entire focusing range and field of view. The optical properties shall conform to the following:

(a) Magnification, minimum - 30 diameters, plus or minus 5 percent.
(b) Effective aperture of objective lens, minimum - 40 mm.
(c) Focus, minimum - not more than 12 feet.
(d) Field of view, minimum - 60 minutes of arc.
(e) Resolution - 4.0 seconds of arc or better.
(f) Eye relief distance - not less than 7 mm.

3.7.1.1 Lens coating. Internal lens-air surfaces, except the glass diaphragm, shall be given a reflection-reducing coating conforming to MIL-C-675.

3.7.1.2 Reticle. The reticle shall consist of a pattern of opaque lines permanently registered on optical glass and mounted on a metal reticle cell. The reticle shall be arranged to eliminate distortion during collimation. Means for achieving collimation adjustments shall be positive and accessible. The reticle cell shall be adjustable for precise positioning of crosslines in the horizontal and vertical planes. Horizontal adjustment shall be by means of 3 or 4 capstan head screws that fasten the cell to the telescope and permit rotation of the cell. The reticle cell shall be provided with peg adjustments, a fitted sliding component, and adjustment screws for movement of the reticle in a vertical plane. Horizontal and vertical stadia lines shall be a minimum of 5 minutes of arc in length, parallel to the corresponding crosslines, and spaced as specified (see 6.2), from the vertical axis of the level to the rod. The rod shall be located at a

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predetermined distance greater than 100 feet or 30 meters from the level. Readings shall not result in errors that exceed 0.3 percent of the horizontal distance and shall not include allowance for the instrument constant. The width of all reticle lines shall be 3 seconds of arc plus 0.0 seconds minus 0.5 seconds.

3.7.2 Eyepiece. The eyepiece shall be so constructed that the eye of an observer can be placed in a position to coincide with the axis of the exit pupil of the telescope. The lens shall be tube-mounted and adjustable by means of a spiral movement mechanism that permits focusing on the reticle pattern and therby achieves parallax-free focusing. The eyepiece shall be equipped with a diopter scale. The scale shall be marked in plus and minus units for use in referencing the setting. The eyepiece shall be free from backlash, and shall provide a uniform focusing motion with sufficient tension to maintain its setting.

3.8 Circular level.

3.8.1 Type I levels. The circular level vial shall have a sensitivity of 10 minutes of arc $\pm$ 2 minutes, per 2 mm division.

3.8.2 Type II levels. The circular level vial shall have a sensitivity of 7 to 15 minutes of arc per 2 mm division. The circular level sensitivity shall be complementary to the range of the compensator; it shall be not less than 75 percent plus 0.0 minus 1.0 minute of arc, of the working range of the instrument compensator for the individual leveling instrument to which it is attached.

3.8.3 Viewer. On Type I and Type II levels, the circular spirit level viewer shall be an observation prism or mirror. The viewer shall be attached to the telescope body of the instrument in a location permitting a view of the bubble that is eye-height, horizontal, and parallel with the line-of-sight.

3.9 Line-of-sight leveling apparatus.

3.9.1 Type I levels. The Type I levels shall be equipped with a telescope level. The telescope level vial shall have a sensitivity of 8 to 10 seconds of arc per 2 mm movement. The ends of the level bubble shall be visible at all temperatures from minus 25° F to plus 120° F when read through the train of prisms and the built-in magnifying eyepiece. Temperature changes or adjustments shall produce no stresses or strains in the vial glass which will incapacitate the operation of the leveling instrument.

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3.9.2 Type II levels. Type II levels shall be equipped with a compensator. The compensator shall be automatic in movement and shall maintain a true horizontal level line-of-sight at all positions of the telescope after the instrument has been initially leveled at each setup station by means of a circular spirit level (see 3.8.2). The compensator shall be removable from the telescope to permit replacement. The compensator shall have working range of not less than ± 15 minutes of arc. The compensator shall have a dampening capability of not more than 5 seconds of time, and shall permit the instrument to perform to the accuracy specified herein throughout the entire operating temperature range.

movement of the compensating prism although it is tilted, and permit unrestricted oscillations in any direction. The amount of tilt shall be the same as the rated working range and shall be both parallel with and perpendicular to the line of sight.

3.9.2.2 Dampening mechanism. The compensator shall be provided with a dampening mechanism to stop oscillation and permit return and stability to the prism within 5 seconds.

3.10 Support. The level shall be mounted on a tribrach. The tribrach shall consist essentially of a baseplate and three leveling screws. The tribrach baseplate shall have at its center a 5/8-inch nut of unified coarse thread series, (5/8-11-UNC-2B), Class 2B, 11 threads per inch, as defined in Handbook H28. The instrument shall rotate on a tapered bearing surface or ball bearing races. All bearings shall move without binding or end-play throughout the temperature range from minus 25° F to plus 120° F.

3.10.1 Leveling screws. The leveling of the instrument shall be accomplished by movement of three leveling screws, which shall center the circular bubble in the vial. Leveling screws shall operate without binding, backlash, or end-play throughout the temperature range from minus 25° F to plus 120° F.

3.11 Azimuth adjustment mechanism.

3.11.1 Type I level. The azimuth adjustment mechanism shall be a free-movement type with a motion-clamping screw and a tangent screw for slow-motion control or precise adjustment.

3.11.2 Type II level. The azimuth adjustment mechanism shall be friction plates or other device to accomplish a positive telescope azimuth movement. A friction locking plate or brake shall permit lateral adjustment and eliminate need for a clamping screw. Slow-motion control or precise adjustment shall be accomplished by means of an endless tangent screw. Left and right control knobs shall operate the tangent screw.

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3.16 Accessories. The level shall be furnished with a lens-cleaning brush, screwdriver, adjusting pin, objective lens dustcap, eyepiece lens cover, and sunshade.

3.16.1 Waterproof cover. The cover shall be light in weight, flexible, and able to withstand the environmental conditions specified herein without damage (see 6.7).

3.17 Tripod. The level shall be provided with a wide frame tripod having fixed or extension type legs, as specified (see 6.2). When applicable, the extension type legs shall conform to MIL-T-14091. The tripod shall be constructed of hardwood that shall not bend, warp, or be affected by changes in temperature or humidity. The tripod shall be furnished complete, and include a tempered-steel shoe on the lower end of each leg, metal head, and head cover. The shoe shall be hardpointed, fastened to the leg, and equipped with a push-down spur. The head, or an adapter ring when furnished, shall be metal, and have threads to accommodate the level baseplate nonmagnetic material as the head, or of high impact plastic; when the level is mounted on the head, the cover shall be attached to the tripod for storage so that the cover is not capable of movement. When the head-protecting cap is removed and the legs are placed together, the overall length of the tripod shall be not less than 66 inches. A binding strap to fasten the legs together shall be attached to one leg.

3.18 Lubrication. Before the level, accessories, and tripod have been assembled, they shall be lubricated, where applicable. The lubricant used shall be capable of protecting all moving parts throughout the environments specified herein.

3.19 Treatment and painting. The portions of the level, tripod, and carrying case normally painted shall be cleaned, treated, and painted in accordance with MIL-T-704. The interior of the telescope and viewing system shall be finished optical black. Exterior color shall be optional. Nonpermissible galvanic couples (dissimilar metals), as defined in MIL-T-704, shall be insulated as specified in MIL-T-704.

3.20 Identification marking. Each level shall be identified in accordance with MIL-STD-130. In addition, the serial number shall be engraved on each level.

3.21 Fungus resistance. The level shall be treated so that it is resistant to the growth of fungi.

3.22 Workmanship. All parts, components, and assemblies of the level, tripod, and carrying case including castings, forgings, molded parts, stampings, bearings, seals, machined surfaces, and welded parts shall

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movement of the compensating prism although it is tilted, and permit unrestricted oscillations in any direction. The amount of tilt shall be the same as the rated working range and shall be both parallel with and perpendicular to the line of sight.

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3.10.1 Leveling screws. The leveling of the instrument shall be accomplished by movement of three leveling screws, which shall center the circular bubble in the vial. Leveling screws shall operate without binding, backlash, or end-play throughout the temperature range from minus 25° F to plus 120° F.

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be clean and free from sand, dirt, fins, sprues, scale, flux, and other harmful extraneous material. All threads requiring movement and all sliding and rotatable parts shall move without binding, backlash, or end play. Clamping actions, when required, shall be positive and, when unclamped, shall not cause drag. Focusing components or assemblies shall retain their focus when finger pressure is released, and the telescope shall not change focus when the level is rotated or transited. No lateral motion shall exist between the telescope and support. Lenses shall be free of scratches, chips, air bubbles between cemented elements, dirt between the glass and coated film, and cloudiness. Optics shall show sharpness of focus and distinct images to provide clear definition. Metal surfacings on the optical components for all prisms and lenses shall not shift. Liquid within the level vial shall be clear, and the vial shall be tight in the housing. Components and assemblies shall show no evidence of forced attachment. Bearing surfaces shall show no scratches or any irregular surface blemishes. Mirrors shall not be tarnished and backing shall be uniform and unspotted. External surfaces shall be free from burrs and sharp edges, and corners shall be chamfered and rounded except when sharp edges and corners are required.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of the inspection requirements specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined and tested in accordance with referenced specifications and standards, as applicable.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) Qualification inspection (see 4.3).
(b) Initial production inspection (see 4.4).
(c) Quality conformance inspection (see 4.5).
(d) Inspection comparison (see 4.7).
(e) Inspection of packaging (see 4.8).

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4.3 Qualification inspection.

4.3.1 Examination. The level shall be examined as specified in 4.6.1. Presence of one or more defects shall be cause for rejection.

4.3.2 Tests. The level shall be subjected to the tests marked "X" in Column 1 of Table I. Failure of any test shall be cause for rejection.

4.4 Initial production inspection. When specified (see 3.3), one or more initial production levels will be selected at random by the Government from the levels being produced by production tooling and will be examined as specified in 4.6.1 and subjected to the tests marked "X" in Column 2 of Table I to determine conformance to the requirements of this specification. The inspection will be performed by the Government at a site selected by the Government. Acceptance of an initial production level shall not exclude the remaining levels from the quality conformance inspection and acceptance provisions specified in Section 4. In addition to any test specified as part of the initial production test, the Government reserves the right to conduct any or all other tests contained in this specification as part of the initial production test, and failure of such additional tests shall have the same effect as failure of those tests specified as initial production tests.

4.4.1 Inspection failure. Failure of an initial production level to meet any requirement specified herein during and as a result of the examination and tests specified in 4.4 shall be cause for rejection of the initial production level and shall be cause for refusal by the Government to continue acceptance of production levels until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the contractor at no cost to the Government on levels previously accepted and produced under the contract. Any deficiencies found as a result of the initial production inspection will be considered prima facie evidence that all levels accepted prior to the completion of initial production inspection are similarly deficient unless evidence to the contrary is furnished by the contractor and such evidence is acceptable to the contracting officer.

4.5 Quality conformance inspection.

4.5.1 Examination. Each level shall be examined for the defects specified in 4.6.1. Presence of one or more defects shall be cause for rejection.

4.5.2 Tests. Each level shall be subjected to the tests marked "X" in Column 3 of Table I. Failure of any test shall be cause for rejection.

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4.6 Inspection procedure.

4.6.1 Examination. The level and associated equipment shall be examined as specified herein for the following defects:

Major

101. Malfunctioning of parts.
102. Parts, components, or accessories missing, damaged, dirty, or not as specified.
103. Materials not as specified.
104. Safety devices or guard missing or inadequate.
105. Controls not as specified.
106. Leveling screws not as specified.
107. Level mounting not as specified.
108. Fungus treatment not as specified.
109. Parts or components not accessible for servicing or maintenance.
110. Workmanship not as specified.
111. Telescope not internal focusing.
112. Focusing of telescope does not move without backlash or binding.
113. Telescope does not have azimuth adjustment mechanism specified.
114. Telescope optical system does not produce sharp image through the entire focusing range or is not free from internal reflections, spherical or chromatic aberration, coma, astigmatism, or distortion.
115. Internal lens-air surfaces not coated with reflection-reducing film.
116. Reticle lines do not appear opaque and of uniform width when viewed through the telescope.
117. Reticle pattern not as specified.
118. Eyepiece does not focus without binding or backlash or does not retain focus.
119. Accessories not as specified.
120. Bearings not as specified.
121. Reticle adjustment screws not as specified (do not remove cover).
122. Compensator not removable from telescope.
123. Compensator working-range not as specified (do not remove cover plate).
124. Compensator suspension construction not as specified.
125. Compensator bearings not as specified.
126. Compensator dampening mechanism does not operate as specified.

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127. Horizontal circle not as specified.
128. Carrying case not as specified.
129. Level vial not as specified.

Minor

201. Treatment, finishing, or painting not as specified.
202. Identification not as specified.
203. Technical publications missing or not as specified.
204. Items to be furnished with each instrument not as specified or missing.

4.6.2 Tests.

4.6.2.1 Test schedule. The test schedule shall be in accordance with Table I.

Table I. Test schedule

Qualification	Init. Prod.	Qual. Conf.	Test	Test Paragraph	Requirement Paragraph
1	2	3	4	5	6
X	X	X	Accuracy and Precision	4.6.2.3	1.2, 3.6
X	X	-	Magnification	4.6.2.4	3.7.1
X	X	-	Effective aperture	4.6.2.5	3.7.1
X	X	-	Minimum focus	4.6.2.6	3.7.1
X	X	-	Field of view	4.6.2.7	3.7.1
X	X	X	Resolution	4.6.2.8	3.7.1
X	X	X	Reticle line weight	4.6.2.9	3.7.1.2
X	X	-	Stadia constant	4.6.2.10	3.7.1.2
X	X	-	Eyepiece, eye relief distance	4.6.2.11	3.7.1
X	X	-	Level vial sensitivity	4.6.2.12	3.8
X	X	-	Compensator test	4.6.2.13	3.9.2
X	X	-	Horizontal circle test	4.6.2.14	3.13
X	X	-	Accuracy, plane-parallel attachment	4.6.2.15	3.12
X	X	-	Rainfall test	4.6.2.16	3.5.3
X	X	-	Operation test	4.6.2.3.1.1	3.6
X	X	-	Sand and dust test	4.6.2.17	3.5.4
X	X	-	Operation and performance test	4.6.2.3.1	3.6
X	X	-	Low-temperature operation	4.6.2.18.2	3.5.1
X	X	-	Low-temperature storage	4.6.2.18.3	3.5.2
X	X	-	Operation test	4.6.2.3.1.1	3.6

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Table I. Test Schedule (Cont'd)

Qualif-ication	Init. Prod.	Qual. Conf.	Test	Test Paragraph	Requirement Paragraph
1	2	3	4	5	6
X	X	-	High-temperature operation	4.6.2.18.4	3.5.1
X	X	-	High-temperature storage	4.6.2.18.5	3.5.2
X	X	-	Operation and performance	4.6.2.3.1	3.6
X	X	-	Vibration test	4.6.2.19	3.5.5
X	X	-	Operation and performance	4.6.2.3.1	3.6
X	X	-	Shock test	4.6.2.20	3.5.5
X	X	-	Operation and performance	4.6.2.3.1	3.6
X	X	-	Accuracy and precision	4.6.2.3	1.2, 3.6

4.6.2.2 Test conditions. The tests shall be performed in the sequence listed in Table I.

4.6.2.3 Accuracy and precision. Establish two benchmarks 100 meters ± 1 meter apart, and at elevations within 1 meter for the test on a strip of level ground. Precise meter rods, which have been calibrated by the National Bureau of Standards, shall be guyed over each benchmark and set vertical with care. The faces of the rods shall be kept from the sun's rays. Tests shall be performed without shelter and at the external climatic conditions existing at the place of the test. Set up the instrument on a line between the two rods at 10 uniformly spaced positions. Readings of both rods are taken from each of the 10 instrument positions. The level shall be first set up near one rod and then at successive positions toward the other rod. The complete test consists of two forward and two backward runs, a total of 40 determinations of the difference in elevation of the two end stations. The rods are read to the nearest thousandth for levels without a plane-parallel attachment, and to the nearest ten-thousandth for levels with the plane-parallel attachment. Determine the accuracy of the levels by the Princeton Standard Test for estimating the accuracy of a level. The levels shall be rated as follows:

ORDER OF ACCURACY	ERROR (SECONDS OF ARC)
a. First Order	1".54 or less
b. Second Order	3".07 or less
c. Third Order	6".14 or less

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The rating of a certain accuracy will result in level lines that will check in 90 percent of the runs if the average length of sight is 60 meters.

4.6.2.3.1 Operation and performance. The level shall be operated and checked as follows to determine its operational capabilities.

4.6.2.3.1.1 Operation.

(a) Manipulate all movable parts and check for freedom of movement and noninterference with other parts in various operating positions throughout their maximum range.
(b) Fit or attach all accessories and attachments to the level and check for functionability.
(c) Check level for free movement to determine if the bubble ends remain visible in the coincidence viewing eyepiece.
(d) Check compensator for free movement to determine if dampening mechanism operates properly.
(e) Rotate the level slowly about its vertical axis and check for uniform and steady operation.
(f) Check all eyepiece movements for uniform and steady movement, for uniform and clear focus throughout maximum range.
(g) Check parts, components, attachments, and accessories for damage.

Evidence that the level, components, attachments, or accessories will not function; or presence of broken, cracked, or shifted parts on the level, attachments or accessories, shall constitute failure of this test.

4.6.2.3.1.2 Performance. The level, with accessories and attachments, shall be tested along a level base. Each level shall perform within the accuracy category determined in 4.6.2.3. The previously established differences in elevation along the level base shall be of the same or higher order of accuracy as the accuracy category being tested. Nonconformance to 3.6 shall constitute failure of this test.

4.6.2.4 Magnification. The magnifying power, in diameters, of the telescope shall be measured by focusing the telescope for parallel light. Without change of focus, a transparent graduated scale shall be placed in front of the objective lens and as near thereto as practicable. A measuring microscope shall be focused on the image of the scale. The magnification of the telescope shall then be determined in diameters as the ratio of a given length of the scale to the length of its image in the exit pupil. Any other method yielding values accurate within 2 percent may be used in lieu of this method. Magnification for the telescope differing from 30 diameters, plus or minus 5 percent, shall constitute failure of this test.

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4.6.2.5 Effective aperture. A scale graduated in millimeters shall be placed in contact with the front of the objective lens. The effective aperture is the diametrical distance of the lens. An aperture of less than 40 mm shall constitute failure of this test.

4.6.2.6 Minimum focus. The telescope shall be focused on a collimator, the reticle of which has been set to simulate 12 feet (3.7 meters), or the telescope shall be set at minimum focus, and a graduated scale shall be moved along the line of sight until the markings are most sharply defined. The distance from this point of the scale measured along the optical axis to the vertical axis of the instrument is the minimum focus. A minimum focus of more than 12 feet (3.7 meters) shall constitute failure of this test.

4.6.2.7 Field of view. The field of view shall be measured pointing the telescope on a target set on a wall not less than 20 feet from the level. The level's horizontal motion is then locked on the target. Targets shall be carefully set at the opposing limits of the field of view along the horizontal and vertical axis of the telescope, using the horizontal and vertical crosswires as guides. The level is removed from the stand and replaced by a 1-second theodolite. The theodolite is carefully leveled and the angle between the targets in the horizontal, then the vertical plane are measured. A field of view of less than 1 degree shall constitute failure of this test.

4.6.2.8 Resolution. Using Pattern 28 and all coarser patterns of the test chart accompanying National Bureau of Standards Special Publication 374, at a distance of 150 feet from the level, check the resolution. Resolution of more than 4 seconds of arc shall constitute failure of this test.

4.6.2.9 Reticle line weight (width). The reticle line weight (width) shall be determined by sighting on a collimator with a target in angular units, or by the use of any other method or target that will yield comparable values. Nonconformance to 3.7.1.2 shall constitute failure of this test.

4.6.2.10 Stadia constant. The instrument shall be mounted on a tripod or stand and leveled, and a graduated rod shall be located at a distance of 100 feet from the level. With the rod in the vertical position and carefully plumbed, the stadia interval shall be read by means of the stadia hairs. The stadia reading shall be multiplied by the specified stadia constant (see 3.7.12). Variation of the stadia distance by more than $\pm$ 0.3 percent of the measured distance shall constitute failure of this test.

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4.6.2.11 Eyepiece, eye relief distance. The level shall be focussed into a collimator, and an exit pupil tester shall be attached to the eyepiece and used to measure the eye relief distance. An eye relief distance of less than 7 mm shall constitute failure of this test.

4.6.2.12 Level vial sensitivity. The sensitivity of the level shall be determined by mounting the level vial on a level trier or by any other method yielding values accurate to within 5 percent. The plane of the graduated vials shall be changed a sufficient amount to cause a change of not less than two graduations in the position of the level bubble. The plane of the ungraduated vials shall be changed a sufficient amount to cause a change of not less than 4 mm in the position of the level bubble, as indicated on a scale placed adjacent to the vials.

4.6.2.12.1 Type I level vials. Nonconformance to 3.8.1 or 3.9.1 shall constitute failure of this test.

4.6.2.12.2 Type II level vials. Nonconformance to 3.8.2 shall constitute failure of this test.

4.6.2.13 Compensator. Place the instrument on a collimator instrument stand, carefully leveled, and alined with the left, center and right targets of the collimator. The instrument shall then be sighted over one of the foot screws at one of the collimator targets. A precise calibrated level shall be placed on the instrument and parallel with the line of sight. by use of the precise calibrated level and one foot screw, the instrument is tilted the applicable angular amount for the compensator being tested. The instrument sighting is then compared to the original sightings on all three collimator targets. Evidence of a change in the line of sight, friction in the compensator's movement, or nonconformance to 3.9.2 shall constitute failure of this test.

4.6.2.14 Horizontal circle. The instrument shall be mounted on a collimator test stand, carefully leveled and sighted on the left target, then the center and right targets of the collimator. A reading of the horizontal circle, to the nearest minute of arc, shall be taken at each pointing on the targets. This procedure shall be accomplished on different parts of the circle. The angles, as determined by the level's horizontal circle, shall be compared to the fixed 90 degree angles of the collimator. Graduation error resulting in reading errors in excess of $\pm$ 1 minute of arc shall constitute failure of this test.

4.6.2.15 Accuracy, optical micrometer - plane-parallel attachment. To test the optical micrometer - plane-parallel attachment, the instrument shall be mounted on a test stand, carefully leveled, and sighted on a precise 4 cm vertical scale. Using the 4 cm vertical scale, the run of the micrometer shall be tested by 5 observations of the scale and a mean run of the micrometer determined. Evidence of an error in the run of the micrometer greater than $\pm$ 0.0001 meter shall constitute failure of this test.

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4.6.2.16 Rainfall test.

4.6.2.16.1 Level. The level shall be placed in a rain-testing chamber not less than 36 inches high, 36 inches wide, and 36 inches deep. The chamber shall be equipped with water spray nozzles to provide a uniform flow of water over the test area at a rate of 3 inches, plus or minus 1 inch, per hour. The direction of rainfall shall be variable up to 45 degrees from the vertical. Water temperature shall be 52° F to 68° F. The level shall temperature-stabilize to within 5° F of the water temperature. The level shall be subjected to this for 5 minutes. Evidence of water in the telescope, in the optical system, in the foot screw dust caps, on bearings, inside the level vial prism housing or compensator housing, or inside the optical micrometer assembly shall constitute failure of this test.

4.6.2.16.2 Carrying case. The level in the carrying case shall be subjected to the test specified in 4.6.2.16.1 for a period of 60 minutes. Swelling or buckling, loose hardware, inability to close, or evidence of water in the carrying case, the telescope, the optical system, the foot screw dust caps, prism housing, compensator housing, or the optical micrometer assembly shall constitute failure of this test.

4.6.2.16.3 Transporting case. The rainfall test of the transporting case may be conducted concurrently with the rainfall test of the carrying case except that the transporting case shall be subjected to the rainfall test specified in 4.6.2.16.1 for a period of 1 hour. Evidence of water in the transporting case, swelling or buckling, loose hardware, or inability to close the case shall constitute failure of this test.

4.6.2.17 Sand and dust tests.

4.6.2.17.1 Level. The level shall be placed in a sand and dust testing chamber not less than 36 inches high, 36 inches wide, and 36 inches deep. The sand and dust of 140 mesh silica flour (see 6.5) shall pass through the chamber blown by a 100 to 500 feet per minute wind. The density of sand and dust shall be maintained at 0.1 to 0.5 gram per cubic foot within the test area. The internal temperature shall be maintained at 77° F, plus or minus 5° F. The relative humidity shall not exceed 30 percent. The level shall be subjected to this test for a period of 15 minutes. Evidence of sand or dust in the telescope, in the optical system, in the foot screw dust caps, on the bearings, inside the level vial prism housing or compensator housing, or inside the optical micrometer assembly shall constitute failure of this test.

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4.6.2.17.2 Carrying case and transport case. The carrying case and transport case, separately, shall be subjected to the test specified in 4.6.2.17.1 for a period of 1 hour. Evidence of sand or dust in the carrying case shall constitute failure of this test.

4.6.2.18 Low-temperature and high-temperature tests.

4.6.2.18.1 Low-temperature and high-temperature test chamber. The low- and high-temperature tests shall be conducted in a test chamber of a size that will house the level mounted on its tripod, the cases, and accessories, and shall permit walking around the periphery of the tripod legs.

4.6.2.18.2 Low-temperature operation. The level shall be mounted on its tripod. The mounted level shall be covered with the waterproof cover. The level, carrying case, transporting case, accessories, and waterproof cover shall be placed in a temperature chamber. The temperature in the chamber shall be gradually lowered to minus 25° F and maintained at that temperature for not less than 6 hours. The level shall then be subjected to the operation test specified in 4.6.2.3.1.1. The waterproof cover shall be folded 4 times, and loss of flexibility or cracking shall constitute failure of the test for the cover. Inability of the instrument and of the accessories to operate at minus 25° F, or nonconformance to 3.5.1 shall constitute failure of this test.

4.6.2.18.3 Low-temperature storage. Upon completion of the test specified in 4.6.2.18.2, the level shall be placed in its carrying case, which shall then be placed in the transporting case. The tripod shall be folded and all accessories stowed for transport. The temperature of the test chamber shall be lowered to minus 50° F and maintained at this temperature for 12 hours. The temperature shall then be gradually raised to that of the laboratory and the operation test of 4.6.2.3.1.1 shall be conducted. Upon completion of the operation test, the level shall be subjected to the performance test of 4.6.2.3.1.2. Nonconformance to the failure criteria specified in 4.6.2.3.1.1 and 4.6.2.3.1.2 shall constitute failure of this test.

4.6.2.18.4 High-temperature operation. The level mounted on its tripod, the carrying case, transporting case, accessories, and waterproof cover shall be placed in a temperature control chamber. The temperature in the chamber shall be gradually raised to 120° F and maintained at this temperature for not less than 4 hours. The level and accessories shall then be submitted to the operation test specified in 4.6.2.3.1.1. Evidence of damage to the waterproof cover or failure of the level or accessories to operate shall constitute failure of this test.

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4.6.2.18.5 High-temperature storage. Upon completion of the test specified in 4.6.2.18.4, the level shall be placed in its carrying case which shall in turn be placed in the transporting case. The tripod shall be folded and all accessories stowed for transport. The temperature of the test chamber shall then be raised to 155° F and the temperature maintained for 4 hours. The temperature shall then be gradually lowered to that of the laboratory and the operation test of 4.6.2.3.1.1 shall be conducted. Upon completion of the operation test, the level shall be subjected to the performance test of 4.6.2.3.1.2. Nonconformance to the failure criteria specified in 4.6.2.3.1.1 or 4.6.2.3.1.2 shall constitute failure of this test.

4.6.2.19 Vibration. The level shall be subjected to vibration testing in accordance with the following.

4.6.2.19.1 Carrying case removed from the transport case. The level in the carrying case shall be rigidly mounted to the vibration table and subjected to two sweeps of sinusoidal cycling vibration in each of the three mutually perpendicular axes. Each sweep shall be 5 to 55 to 5 Hertz and traversed in 15 minutes. The vibration amplitude shall be 0.10-inch double amplitude from 5 to 17 Hz and 1.5 g from 17 to 55 Hz. Upon completion of this test, the level shall be mounted on its tripod and tested for operation as specified in 4.6.2.3.1.1. Evidence of damage to the theodolite or the case shall constitute failure of the test.

4.6.2.19.2 Transport case. The level in the carrying-transport case shall be tested in accordance with MIL-STD-810, Procedure X, Table 514.2-VII, Figure 514.2-7, Curve AW with the exception that only five sweeps of vibration cycling shall be performed for a total of 1 hour of vibration in each axis. Upon completion of this test, the level shall be mounted on its tripod and tested for operation as specified in 4.6.2.3.1.1. Evidence of damage to the level or the case shall constitute failure of the test.

4.6.2.20 Shock. The level in the carrying case shall be flat dropped in an upright position from a height of 6 inches onto a level wood floor. The level shall then be subjected to the operation and performance test specified in 4.6.2.3.1. The level in its carrying case shall then be placed in the transporting case and, after being secured for shipment, shall be dropped from a height of 12 inches so that one bottom edge of the transporting case strikes a level concrete floor when the bottom of the transporting case is inclined 15 to 20 degrees from the horizontal. The performance. Evidence of damage to the level or the cases, or nonconformance to the failure criteria of 4.6.2.3.1.1 or 4.6.2.3.1.2 shall constitute failure of this test.

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4.6.2.20.1 Shock test of transporting case. A dummy load in the configuration of the carrying case and equal in weight to the level and the carrying case, shall be placed in the transporting case. The transporting case shall be closed and secured as for shipment and subjected to eight drops, one on each bottom edge and one on each bottom corner onto a horizontal concrete floor from a height of 42 inches. The height of drop shall be measured from the concrete floor to the nearest edge or corner of the transporting case when suspended prior to each drop. The presence of splinters or cracks; broken, cracked, or loose hardward; or opened joints, shifted padding, or deformation of the transporting case shall constitute failure of this test.

4.7 Inspection comparison. The Government may select levels at any time during the contract production period and subject these levels to the examination and tests specified in 4.4 to determine conformance to the requirements of this specification. The inspection will be performed by the Government, at a site selected by the Government, on levels selected at random from those which have been accepted by the Government and will not include the previously inspection preproduction model and initial production levels. In addition to any test specified as part of the inspection comparision, the Government reserves the right to conduct any and all other tests contained in this specification as part of the inspection comparison, and failure of such additional tests shall have the same effect as failure of those tests specified as inspection comparison.

4.7.1 Inspection failure. Failure of an inspection comparison level to meet any requirement specified herein during and as a result of the examination and tests specified in 4.7 shall be cause for rejection of the inspection comparison level and shall be cause for refusal by the Government to continue acceptance of production levels until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the contractor at no cost to the Government on levels previously accepted and produced under the contract. Any deficiencies found as a result of the inspection comparision will be considered prima facie evidence that all levels accepted prior to the completion of inspection comparison are similarly deficient unless evidence to the contrary is furnished by the contractor and such evidence is acceptable to the contracting officer.

4.8 Inspection of packaging. The preservation, packing, and military marking shall be examined to determine compliance with the quality assurance provisions of the referenced documents in Section 5 of this specification. The commercial marking shall be examined to determine compliance with the requirements in Section 5 of this specification.

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5. PACKAGING

5.1 Preservation. Preservation shall be Level A or B or Commercial as specified (see 6.2).

5.1.1 Level A. The level with all its associated components shall be preserved in accordance with the Level A preservation and packaging requirements of MIL-S-3534. When a transport case is furnished, and Level A preservation is specified, the transport case shall be considered the outer container in lieu of the fiberboard box specified in MIL-S-3534.

5.1.2 Level B. The level with all its associated components shall be preserved in accordance with the Level B preservation and packaging requirements of MIL-S-3534. When a transport case is furnished and Level B preservation is specified, the transport case shall be considered the outer container in lieu of the fiberboard box specified in MIL-S-3534.

5.1.3 Commercial. The level with all its associated components shall be preserved in a manner to assure protection against deterioration and damage from the supply source to the initial destination.

5.2 Packing. Packing shall be Level A or B or Commercial as specified (see 6.2).

5.2.1 Level A. Each complete level, preserved as specified in 5.1, shall be packed in accordance with the Level A packing requirements of MIL-S-3534.

5.2.2 Level B. Each complete level preserved as specified in 5.1, shall be packed in accordance with the Level B packing requirements of MIL-S-3534.

5.2.3 Commercial. Each complete level, preserved as specified in 5.1, shall be packed in a manner to assure carrier acceptance and safe delivery to destination at lowest ratings in compliance with Uniform Freight Classification rules or National Motor Freight Classification rules.

5.3 Marking.

5.3.1 Military. Marking for military levels of protection shall be in accordance with MIL-STD-129.

5.3.2 Commercial. Marking for commercial packaging shall reflect, as a minimum, the item name, National Stock Number, quantity, contract or purchase order number and appropriate address markings on the shipping container. The marking shall be applied with a waterproof material, be legible, and shall provide a definite contrast.

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(a) Title, number, and date of this specification.

6. NOTES

6.1 Intended use. Type I and Type II levels are intended for 1st and 2nd order leveling. Levels will be used according to the accuracy category.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type, style, and category of level required (see 1.2).
(c) When an optical micrometer and plane-parallel attachment is required, and type of image required (see 3.1.3).
(d) When an initial production level is required and number required (see 3.3).
(e) Stadia interval required (see 3.7.1.2).
(f) When transport case is required (see 3.15).
(g) Type of tripod legs required (see 3.17).
(h) Degree of preservation and packing required (see 5.1 and 5.2).

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time set for opening of bids, qualified for inclusion in the applicable Qualified Products List, whether or not such products have actually been so listed by that date. The attention of the contractor is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the qualified products list is the Commander, U.S. Army Mobility Equipment Research and Development Command, ATTN: DRXFB-RE Fort Belvoir, VA 22060, and information pertaining to qualification of products may be obtained from that activity.

6.4 Classification relationship. The relationship between classification of the levels included in this revision of the specification and the superseded documents is as follows:

MIL-L-	MIL-L-3841B	MIL-L-28590(YD)
Type I	Yes	None
Type I, Style A	Yes	None
Type I, Style B	None	None
Type II	None	Yes
Type II, Style A	None	Type I
Type II, Style B	None	Type II

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6.5 Sand and dust. The sand and dust to be used in the sand and dust test is commercially known as "140-mesh silica flour". 140-mesh silica flour produced by the Fenton Foundry Supply Company, Dayton, OH 45403, and Ottawa Silica Company, Ottawa, IL 61350, or equal will be satisfactory.

6.6 Definition.

6.6.1 Damage. Damage is deformation, corrosion, loosening of parts, malfunction, breakage, or change of fit of any component or part, or any physical condition resulting in noncompliance of the level with the provisions of this specification.

6.7 Data requirement. The contracting officer should include requirements for such data as technical publications, instructional materials, illustrated parts lists, and contractor's maintenance and operation manual to be furnished with each level.

6.8 Provisioning. The contracting officer should include provisioning requirements for repair parts and maintenance tools as necessary (including any special tools), and instructions regarding shipment of levels.

Custodians:
Army - ME
Navy - YD
Air Force - 99

Review activity:
Army - MI

Preparing activity:
Army - ME

Project 6675-0473

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

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DOCUMENT IDENTIFIER AND TITLE

MIL-L-52918 Levels, Surveying, Self-Leveling and Tilting

NAME OF ORGANIZATION AND ADDRESS	CONTRACT NUMBER
	MATERIAL PROCURED UNDER A ☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*	TELEPHONE NO.
	DATE

DD FORM 1 JAN 72 1426

REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

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